

SatQuery AI - In Simple Terms 🧩

What is This Project?

text

Think of it like this:

Google Maps lets you ask:

"Show me restaurants near me"

And it shows you the answer visually

SatQuery AI lets anyone ask:

"Show me flooded areas in Bihar"

And satellite images answer you visually

The Real Problem Today

text

RIGHT NOW:

Satellite images exist for EVERYTHING

Every flood → Captured by satellite

Every forest fire → Captured by satellite

Every new building → Captured by satellite

Every crop change → Captured by satellite

BUT:

To USE these images you need:

✗ PhD in Remote Sensing

✗ Know GIS software (QGIS, ArcGIS)

✗ Understand satellite data formats

✗ Know which model to run

✗ Months of training

So WHO suffers?

District Collector during flood

→ Cannot quickly see affected villages

Forest officer tracking deforestation

→ Cannot compare old vs new images easily

Farmer needing crop health info

→ Cannot access satellite insights

NDRF rescue team

→ Cannot quickly map disaster zones

What SatQuery AI Does

text

BEFORE SatQuery AI:

Expert uploads satellite image

↓

Opens complex GIS software

↓

Runs multiple specialized programs

↓

Waits hours for analysis

↓

Gets technical output

↓

Translates to simple answer

↓

Total time: Days

AFTER SatQuery AI:

Anyone uploads satellite image

↓

Types simple question in English

↓

Gets visual answer with explanation

↓

Total time: 30 seconds

Three Things It Can Do

Thing 1: Answer Questions About One Image
text

You upload ONE satellite image

You ask:

"What do you see in this image?"

It answers:

"This image shows agricultural land covering 60% of the area, with a river running through the eastern side and a small town in the north"

Simple. Like asking ChatGPT but for satellite images.

Thing 2: Compare Two Images Over Time
text

You upload image of SAME place from TWO different years

2019 image + 2024 image of Mumbai

You ask:

"What changed in 5 years?"

It answers:

"Built-up area increased by 23%

Forest cover reduced by 15%

New highway constructed in north

Water body reduced by 8%"

+ Shows you a COLOR MAP of exactly where changes happened

Like a before-after comparison but intelligent and automatic

Thing 3: Use Two Types of Cameras Together
text

Satellites have TWO types of cameras:

Camera Type 1 (Optical):

→ Like a normal camera

→ Gives color photos

→ Cannot see through clouds

→ Good for land, buildings, water

Camera Type 2 (SAR/Radar):

→ Like a bat using sonar

→ Works through clouds

- Works at night
- Good for structural info

SatQuery AI uses BOTH together:

"Find all water bodies and buildings in this area"

System:

- Uses optical for color (water = blue)
- Uses SAR for structure (buildings)
- Combines both answers
- Gives you complete picture

Like having two expert scientists working together to answer you
The "Smart" Part - The Agent text

The BRAIN of SatQuery AI works like a smart assistant at a company:

You (CEO) ask a question

↓

Smart Assistant (Agent) receives it

↓

Agent thinks:

"This question needs:

- Expert 1 for image analysis
- Expert 2 for change detection
- Expert 3 to combine answers"

↓

Agent calls right experts

↓

Agent combines their answers

↓

Agent gives you ONE clean answer
+ Shows what it did step by step

You never talk to the experts directly
Agent handles everything for you

This is exactly what Yash built in CREVA with 35 agents
Just applied to satellite images
Who Uses This and How text

USER 1: NDRF Officer During Flood

Uploads satellite image of flooded area

Types: "Which villages are underwater?"

Gets: Map with highlighted villages

- + List of affected areas
- + Estimated flood coverage

Action: Sends rescue teams accurately

USER 2: Forest Department Officer

Uploads 2020 and 2024 forest images

Types: "How much forest was cut?"

Gets: Exact percentage lost

- + Map showing exact locations
- + Year-wise comparison

Action: Files legal case with evidence

USER 3: Agriculture Ministry

Uploads current crop satellite image

Types: "Which areas have poor crops?"

Gets: Highlighted stressed crop regions

+ Severity assessment

+ Area calculation

Action: Sends targeted farmer support

USER 4: Urban Planner

Uploads city satellite image

Types: "Where is illegal construction?"

Gets: Highlighted new structures

+ Comparison with approved maps

Action: Sends inspection teams

Why ISRO Cares About This

text

ISRO launches satellites every year

India has 50+ operational satellites
generating massive amounts of data

Problem:

99% of this data sits unused

because only experts can read it

SatQuery AI =

Opens ISRO's data to ALL of India

Any government officer

Any farmer

Any researcher

Any student

Can now USE satellite data

Just by asking a question

ISRO's dream:

"Democratize space data for India"

Your project:

Makes that dream real

In ONE Sentence

text

SatQuery AI is like ChatGPT
but for satellite images -
anyone can ask questions about
any place on Earth and get
intelligent visual answers
from real satellite data
without needing any
technical knowledge.

Why You Will Win With This

text

Every other team will build:

- Apps
- Chatbots
- Dashboards

You will build:

- Something that talks to satellites

When judges ask "what does it do?"

Other teams say:

"It automates government processes"

You say:

"Sir, type any question.

I will show you the answer
from satellite images of India.
Right now. Live."

Types: "Show flood damage in Assam 2024"

System shows satellite map with answer

That moment = you win.